2008 JC2 PRELIMINARY EXAMINATIONS
H2 Biology
Paper 2 (Core)
MARK SCHEME

Disclaimer
This mark scheme should be used purely as a revision tool.
Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Key to mark scheme

ref.	with reference to
/	alternative wording
(brackets)	recommended wording that is optional
underline	accept this word only
OVP	other valid points

Paper 1 (MCQ)

Q1	A	**Q2**	B	**Q3**	B	**Q4**	C	**Q5**	D
Q6	C	**Q7**	C	**Q8**	B	**Q9**	C	**Q10**	B
Q11	D	**Q12**	C	**Q13**	C	**Q14**	D	**Q15**	D
Q16	D	**Q17**	B	**Q18**	C	**Q19**	D	**Q20**	B
Q21	B	**Q22**	D	**Q23**	A	**Q24**	A	**Q25**	A
Q26	A	**Q27**	D	**Q28**	C	**Q29**	D	**Q30**	C
Q31	B	**Q32**	C	**Q33**	D	**Q34**	A	**Q35**	D
Q36	C	**Q37**	C	**Q38**	C	**Q39**	A	**Q40**	D

Paper 2 (Core)

Section A

QUESTION 1 [Total: 12]

In humans a gene coding for a certain protein is located on chromosome 12. The gene contains 1125 base pairs, but the mature mRNA strand only contains 456 bases. The protein contains 140 amino acids.

Fig. 1.1 shows seven triplets that code for this protein.

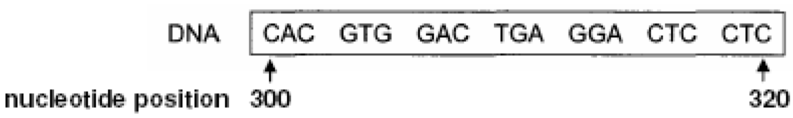


Fig. 1.1

(a) (i) State the total number of bases that code for the amino acids in this protein? [1]

140 aa x 3 bases = 420 bases;

(ii) Using your answer in (a)(i), explain the number of bases in the gene and the mature mRNA. [2]

1 presence of introns;
2 spliced from pre / primary mRNA to form mature / secondary mRNA;

(b) Using Table 1.2, determine the amino acid sequence for the portion of DNA given in Fig. 1.1. [1]

val – his – leu – thr – pro – glu – glu

Two mutations are commonly associated with this gene on the 20th nucleotide; a single base deletion at position 315 or a single base substitution by adenine at the same position.

(c) (i) Describe the structure of a polynucleotide. [3]

1 *ref*. polymer of deoxyribonucleotides;
2 with deoxyribose, phosphate + nitrogenous base;
3 linked by phosphodiester bonds;

(ii) Explain which of the above mutation is likely to be more harmful . [3]

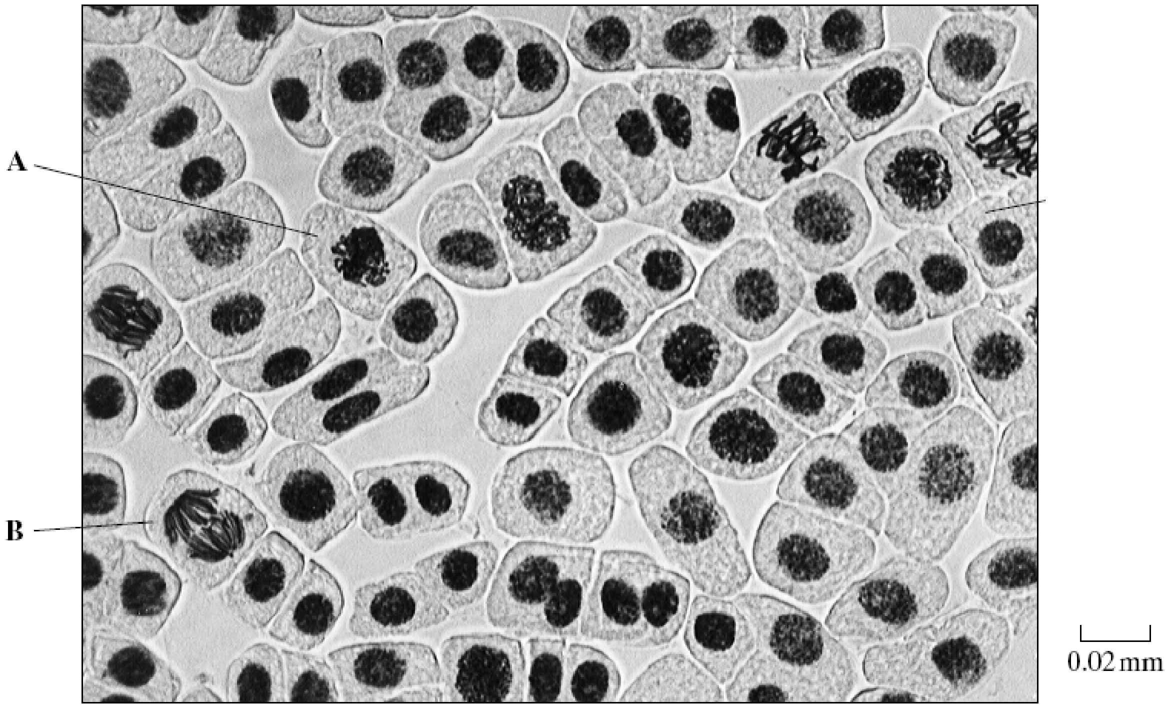
Deletion
1 single base deletion leads to frameshift;
2 alters aa seq downstream of mutation;
3 diff R grps, diff bonds, diff folding, diff 3D config;
4 *ref*. change in structure leading to a change in function
or
Substitution
5 sub by A results in stop codon;
6 truncated / shortened polypeptide;
[pt. 1 accepted with pts 5-6]

(iii) Explain how the change in sequence of the DNA molecule can affect the phenotype of the organism . [2]

1 change in aa seq leads to change in protein 3D config;
2 protein could be an enz / receptor / channel / *etc*. involved in metabolic rxns / recognition / movement of substances / *etc*. resulting in a phenotype,
3 non-functional protein leading to a change in phenotype;

QUESTION 2 [Total: 9]

(a) The photograph shows cells from an onion root tip. The root tip has been squashed and stained to show the stages of mitosis.

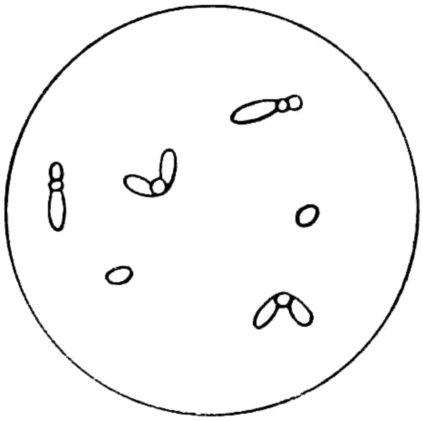


(i) At what stage of mitosis is cell A? [1]
Prophase

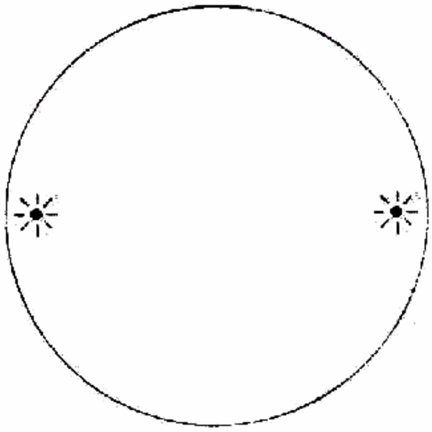
(ii) What is the evidence that cell B is in anaphase? [1]

1 Sister chromatids, now called <u>chromosomes</u>, have moved / pulled apart to opposite poles /ends of cell
2 <u>Distinctive 'v' shape of sister chromatids</u>, now called chromosomes, observed at opposite ends of cell;.

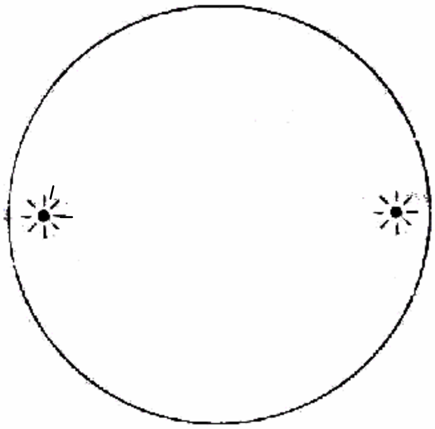
(b) The diagram below represents the chromosomes contained in a somatic cell of an animal (2n= 6).



Using the same chromosomes shown, complete the empty cells below to show clearly the essential differences between metaphase of mitosis and metaphase I of meiosis. [3]



Metaphase (Mitosis)



Metaphase I (meiosis)

Alignment in (mitosis) [1]
Pairing of homologous chromosomes (meiosis) [1]
Chromosomes drawn consisting of sister chromatids + correct size. [1]

(c) Explain why haploid cells need to be produced during a lifecycle which includes sexual reproduction. [2]

1 need to maintain constant/normal number of chromosomes (for individual/species);
2 fusion of gametes / haploid cells restores diploid condition / no.;
3 prevent doubling of chromosomes if diploid gametes fuse;

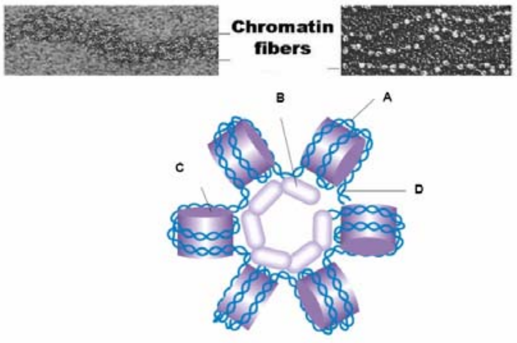
R: restore diploid cell

(d) Explain how cancer is a result of uncontrolled cell division. [2]

1 proto-oncogenes mutate / presence of oncogenes to enhance / stimulate cell division;
2 mutant tumour suppressor genes unable to inhibit / cannot cause cell death / unable to stop cell cycle
3 Telomerase activity not lost so cell continue to divide

QUESTION 3 [Total: 12]

The following diagram shows the DNA packing of chromatin fibres in a eukaryotic cell.



(a) Describe how DNA is organized to form chromosomes. [5]

1 ionic /electrostatic bonds / attraction between histones and DNA (to form a nucleohistone complex);
2 formation of a nucleosome when DNA winds (twice) around an histone (octamer);
3 formation of a 30nm chromatin fibre / solenoid when 6 nucleosomes form higher order helix / fold further;
4 formation of looped domain / radial loops;
5 ref. to loop domains coil and fold further to compact / condense (all the chromatin into chromosomes);

[***R**: DNA condense together*]

(b) Explain briefly how simultaneous modifications of DNA and histones can control gene expression. [2]

1 DNA methylation and histone deacetylation can repress transcription/converse for promotion of transcription;
2 methylation of DNA turns off gene expression;
3 histone deacetylation prevents transcription enzymes from having access to genes;

* must have correct combination (i.e. mythylation and deacetylation both, for switching off gene) (otherwise max 1m)
* R: transcription machineary in pt 3

Gene expression in eukaryotes requires the process of transcription and translation.

(c) State how 2 such post transcriptional modifications may be carried out and explain the purpose of each modification. [4]

1 5' capping of transcription with modified guanine nucleotide / 7-methylguanosine cap;
2 protect mRNA from degradation by hydrolytic enzymes/contains important sequences for ribosome attachment;

3 3' end of transcript has a poly A tail;
4 inhibits degradation of the RNA / help ribosome attachment / facilitate export of mRNA into the cytosol from the nucleus / increase stability & half-life;

5 RNA splicing where introns of removed from the pre-mRNA transcript
6 single gene to encode more then one kind of polypeptide (alternative splicing)/so that coding sequences code for the correct polypeptide chain to be synthesized / continuous coding sequence;

(d) Suggest a type of post-translational modification and state why it may be necessary. [1]

1 biochemical modification, e.g. glycosylation / phosphorylation / cleavage to yield functional protein;
2 protein degradation (by proteosomes) to remove ubiquinated protein when it is not required

QUESTION 4 [Total: 9]

(a) (i) Draw a genetic diagram to show the gametes and the genotypes and phenotypes of the F_1 and F_2 generations of the two crosses. Give the ratio of phenotypes in the F_2 generation. [5]

Parental phenotypes:	White flowers				x	White flowers			
Parental genotype:	CCpp					ccPP			
Parental gametes:	Cp					cP			
F_1 genotypes:					CcPp				
F_1 phenotypes:	All purple flowers								
Intercrossing F_1:	CcPp				x	CcPp			
F_1 gametes:	CP	Cp	cP	cp		CP	Cp	cP	cp

Punette square showing F_2 genotypes

	CP	**Cp**	**cP**	**cp**
CP	CCPP	CCPp	CcPP	CcPp
Cp	CCPp	CCpp	CcPp	Ccpp
cP	CcPP	CcPp	ccPP	ccPp
cp	CcPp	Ccpp	ccPp	ccpp

Legend

Purple flowers
White flowers

F_2 genotypes:
F_2 phenotypes: Purple flowers White flowers
F_2 phenotypic ratio: 9 7

1 parental gametes – Cp and cP
2 F_1 genotypes – CcPp
3 F_1 gametes – CP, Cp, cP, cp

4 Punett square with legend / F_2 genotypes corresponds phenotypes
5 9 purple flowers : 7 white flowers

(ii) Suggest the biochemical basis for this form of epistasis. [2]
1 each dominant allele produces an enzyme/protein
2 that controls a step in the synthesis of anthocyanin from a biochemical precursor
3 if a dominant allele is not present (or if both alleles of the gene present are recessive), its step in the biosynthetic pathway is blocked and anthocyanin is not produced

(b) (i) State the calculated value of χ^2 applicable to these data. [1]

5.19 *(3 s.f.)*

Working:

Classes	*Observed number (O)*	*Expected ratio*	*Expected number (E)*
Purple flowers	114	9	99
White flowers	62	7	77
Total	176	16	176

- No. of degrees of freedom = c – 1
  = 2 – 1 = 1

- $\chi^2 = \Sigma \dfrac{(O-E)^2}{E}$

  $= \dfrac{(114-99)^2}{99} + \dfrac{(62-77)^2}{77}$

  = 5.19

(ii) Use the χ^2 table provided to find the probability of the observed ratio departing significantly from the expected ratio. [1]

Probability = 0.02 to 0.05

or 2% to 5% (* must be a range)

Explanation: From the χ^2 table for d.f =1, the calculated χ^2 value of 5.19 lies between 3.84 and 5.41, which represents the probabibilities 0.05 and 0.02, respectively.

In order words, we can say that there is a 0.02 to 0.05 probability that there are significant differences between the observed and expected numbers.

degrees of freedom	probability, p (Probability that chance alone could produce the difference)				
	0.10	*0.05*	*0.02*	*0.01*	*0.001*
1	2.71	**3.84**	**5.41**	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

NOTES:
If p< 5%, that means there is a low probability that it is *chance* that produce the difference. Hence, difference is significant. Reject null hypothesis.

If p>5%, that means there is a high probability that it is *chance* that produce the difference. Hence, difference is not significant. Accept null hypothesis

QUESTION 5 [Total: 13]

Type 2 diabetes is a disease condition that is usually caused by the failure of target cells to respond to insulin. This is due to a defect in the insulin response mechanism of the target cells.

The figure below shows the normal sequence of cell signaling events when insulin reaches its target cell and binds to an insulin receptor.

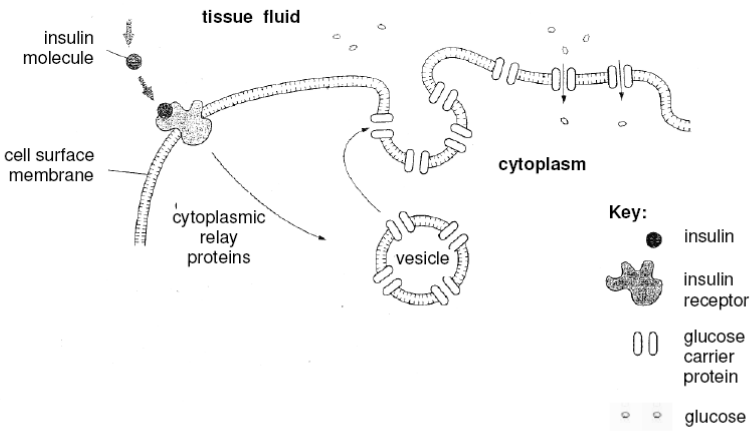


Fig 5.1

(a) State the source of the insulin molecule in the figure above. [1]

1 β-cells of the islets of Langerhans in the pancreas

The structure of the insulin receptor is shown in the figure below.

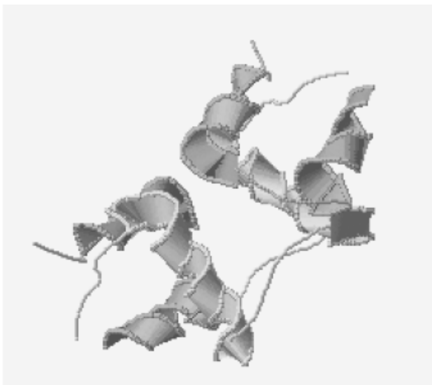


Fig. 5.2

(b) With reference to Fig. 5.1 and 5.2 above,

(i) explain how the binding of insulin to its receptor results in an increase in the phosphorylation of serine and threonine residue side-chains of the cytoplasmic relay proteins. [3]

1 binding causes 2 receptor subunits / polypeptide chains / monomers / molecules to form a dimer / dimerise;
2 activation of tyrosine kinase region leads to (auto) phosphorylation of tyrosine residues on the tail of subunits
3 relay protein(s) bind to specific phosphorylated tyrosine(s) and becomes activated;
4 triggers transduction pathway(s) / phosphorylation cascade to activate serine and threonine kinases / inhibition of serine and threonine protein phosphatases;

(ii) Describe the events subsequent to the phosphorylation of serine and threonine residues of cytoplasmic relay proteins that lead to an increase in the number of glucose channels in the cell surface membrane [2]

1 phosphorylation of cytoplasmic relay proteins constitute a transduced / amplified chemical signal
2 causing movements of vesicles carrying glucose carrier proteins/transporters to move towards cell surface membrane
3 fusion of membrane of vesicles to cell surface membrane and incorporation as channels (*ref*. increasing no. of channels)

The table below shows the relative levels of glucose and insulin in Mr Keanu Rives and Mr Brad Pity. Data was obtained for two conditions, during a 6 hour fast and immediately after a meal high in carbohydrates.

Mr Rives is medically normal with respect to his blood glucose levels. However, Mr Pity has abnormally high levels of blood glucose and has thus been diagnosed as a sufferer of diabetes.

	Blood Glucose Levels / Mg per 100ml	Insulin levels / arbitary units
Mr Rives		
- Fasting	90	5
- After a meal	120	50
Mr Pity		
- Fasting	140	5
- After a meal	190	10

Table 5.3

(c) With reference to Table 5.3, explain why Mr. Pity is considered to be suffering from diabetes. [2]

1 his insulin levels are very low at 10 arb. units after a meal compared to Mr Rives at 50 arb. units.
2 his blood glucose levels are elevated (140-190 mg/100ml) compared to Mr Rives (90 – 120mg/100ml)

(d) Suggest reasons for the following:
(i) Excess blood glucose may cuase frequent thirst. [2]

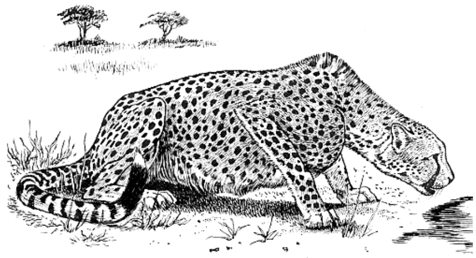
1 water moves out of the cells / tissue fluid into the blood / loss of water from cells by osmosis;
2 cells are deficient in water & demand for water increases;

(ii) Sufferers of diabetes tend to lose weight. [2]

1 diabetics have no glycogen reserves / stores (in liver and muscles) due to lack of insulin;
2 will burn up fat reserves whenever glucose in blood is used up (due to exercise / extended fasting)

QUESTION 6 [Total: 13]

The African cheetah, *Acinonyx jubatus* is an atypical member of the cat family (Felidae). It was suggested that the species had somehow lost its genetic variation. Physiological studies revealed that the populations of cheetahs are so closely related to each other such that skin grafts from one cheetah to another would not provoke immune response, thus suggesting an extreme population bottleneck in the past.



(a) State the genus to which the cheetah belongs. [1]
Acinonyx

(b) Explain what is meant by a bottleneck event and how it can result in reduction in variation. [3]

1 an evolutionary event that resulted in the significant (more than 50%) reduction of a population size;
2 reduction in/small gene pool due to very few individuals / small population of survivors whose genes may not be representative of the parental population;
3 genetic drift occurs in the small population lead to major change in allelic frequency / reduction in genetic variation that occurs entirely due to chance;
4 rarer alleles lost from the population due to genetic drift;

(c) Suggest two problems that could result from the genetic uniformity faced by the South African cheetah population. [2]

1 low no. of alleles reduce probability of a favourable characteristic in the population;
2 changes in environment could result in widespread death of the species / species extinction / reduce chances of survival; (**R**: mention of predators only, unless "the inability to hunt well" is mentioned together)
3 countinuous inbreeding between closely related individuals could lead to inbreeding depression / reduced fitness;

(d) Suggest and explain how variation in the cheetah population can increase slowly but erratically. [2]

1 random mutation resulting in new alleles that increases genetic variation;
2 erratic because natural selection will select for the better adapted individuals with the mutation;
3 because of small gene pool / population with higher chances of inbreeding that would take a long time to increase variation;

(e) Describe how the molecular clock can be used to date evolutionary events such as how long since the cheetah species and the domestic cat species have diverged. [3]

1 Based on the assumptions that neutral mutations are fixed at a relatively constant rate and used as molecular clock;
2 compare DNA sequence of a particular common gene (between the cheetah and domestic cat);
3 determine rate of evolutionary change in another gene / DNA sequence;
4 number of mutation / substitution / difference in genetic sequence is used to calculate the length of time since divergence;

QUESTION 7 (Total = 12 marks)

(a) State the <u>precise</u> <u>location</u> of the four stages. [2]

Respiratory Stage	**Location in cell**
Glycolysis	Cytoplasm/ cytosol;
Link reaction	Mitochondrial matrix
Krebs cycle	Mitochondrial matrix *R: Mitochondrion / matrix alone*
Oxidative phosphorylation	Inner mitochondrial membrane / cristae / stalked particles;

2-3 correct: 1mark
1 correct: 0 mark

(b) Explain the results of the graph

(i) Between points A and B [4]

1 unable to make ATP;
/ ADP limiting / no ADP added;
2 oxidative phosphorylation/<u>aerobic</u> respiration, unable to take place;
3 no O_2taken up / no change in O_2 concentration / remains constant;
4 *ref*. to glycolysis can proceed still to convert glucose to pyruvate;

(ii) Between points B and C [3]

1 ATP formed / by chemiosmosis / by using energy built up from ETC;
2 O_2 level decreases;
3 <u>aerobic</u> respiration / oxidative phosphorylation, proceeds;
4 but slows down as ADP is used up;

(iii) After point D [3]

1 <u>aerobic</u> respiration/oxidative phosphorylation, not taking place;
2 glucose / substrate used up;
3 oxygen level remains constant/same, does not decrease further;
4 ADP will not be used up (for chemiosmosis);

Section B

QUESTION 8 [Total: 20]

(a) Contrast the mode of entry of influenza virus and HIV [max 4]

	Influenza virus	**HIV**
Similarities	▪ Enveloped virus (1/2 m)	
Susceptible host cell	▪ Cells in the respiratory tract	▪ T-helper cells, monocytes, macrophages
Viral specificity is mediated by	▪ Sialic acid (host) ▪ Hemagglutinin (viral protein)	▪ CD4 receptor / co-receptor(host) ▪ GP 120 (viral protein)
Mode of entry	▪ Enters by <u>endocytosis</u>. ▪ Host cell membrane invaginates and pinches off forming an endocytic vesicle. ▪ Genome is released when viral envelope fuses with endocytic vesicle.	▪ Enters by <u>fusion</u> of membrane envelope with host cell membrane. ▪ The fusion releases the remainder of the virion into the cytoplasm. ▪ Genome is released and HIV enters the cell.

(b) Describe in detail the reproductive cycle of T4 phage. [9]

Adsorption
1 (attachment sites on phage) recognize and binds/ absorbs to receptors on the bacterium cell wall

Penetration
2 injects genome into the bacterium's cytoplasm

Replication
3 replication of genome by <u>breaking down</u> host DNA to synthesize viral DNA;
4 makes use of host cell DNA polymerase;
5 hijacks host cell <u>transcriptional</u> machinery / makes use of host cell RNA polymerase;
6 use host cell ribonucleotide to make viral mRNA;

7 hijacks protein synthesis machinery / use host cell ribosomes;
8 produce viral enzymes and new virus structure components;

Maturation
9 assembly of phage components;

Release
10 phage-coded <u>lysozyme;</u>
11 breaks down bacterial peptidoglycan cell wall;
12 causing osmotic lysis / lyse bacteria host;
13 release newly assembled phages;

Reinfection
14 reinfection;

(c) With reference to a named virus and its reproductive cycle, outline how specialized transduction could result in a stable transfer of genes into a recipient bacterium. [7]

1 adsorption and penetration into the host cell;
2 forms a prophage
/integrates λ DNA into the host cell chromosome at insertion sites;

3 in response to environmental stimuli / bacteria cell experiences stress;
4 prophage excised from host chromosome
5 host bacterial genes adjacent to either side of the prophage insertion sites may be excised with the prophage
6 and then packaged into a capsid / phage head
7 host cell is <u>lysed</u> and phage λ progeny released

8 progeny λ phage injects the DNA hybrid into recipient cell
9 lysogenization of recipient can occur resulting in the stable transfer of donor genes
/ homologous recombination may occur
/ newly introduced bacteria genes replaces homologous genes on the recipient chromosome.

QUESTION 9 [Total: 20]

9(a) Outline the roles of calcium, sodium and potassium ions in the transmission of impulses. [10]

Calcium (max 2)

1 arrival of nerve impulse at presynaptic knob results in influx of calcium;
2 synaptic vesicles move towards & fuse with presynaptic membrane;
3 release neutrotransmitter into synaptic cleft;
4 impt in aiding the transfer of electrical signal across the synaptic cleft between 2 neurons;

Sodium (max 5)

5 stimulation at / above threshold, permeability of Na+ increases;
6 influx of Na diffuse rapidly into membrane due to opening of voltage-gated Na+ channels;
7 depolarising membrane from -70mV to +40mV;
8 reversal of membrane potential results in action potential;
9 the basis in the transmission of an impulse;

10 neurotransmitter binds with specific receptors on posynaptic membrane;
11 ligand-gated Na+ ion channels open to allow influx of sodium ions into the postsynaptic membrane;
12 local depolarisaion results in EPSP;
13 impt in initiating a new action potential in the post-synaptic neuron;

14 at node of Ranvier allows influx of Na+;
15 leading to depolarization of membrane from -70mV to +40mV;
16 localised reversal in membrane potential for rapid propagation of neural impulse;

Potassium (max 3)

17 permeability of sodium ions decreases / once membrane potential of +40mV is reached;
18 K+ voltage gated channels open for efflux of K+;
19 Down an electrochemical gradient until membrane is reploarised;
20 Impt in restoring membrane potential;
21 to allow new action potential to be initiated;

9(b) Contrast the transmission of nerve impulse along a neurone and synaptic transmission. [6]

Transmission of nerve impulse along a neurone	Synaptic transmission
Electrical in Nature	Chemical in nature
Takes place / travels along one neurone	Involves 2 adjacent neurones
No calcium ions req'd	Calcium ions req'd for triggering the release of neurotransmitter into synaptic cleft;
Only exitatory	Synapses can be excitatory or inhibitory
Depolarisation is due to an influx of sodium ions resulting from an increase in the permeability of axon membrane to sodium ions	Excitatory synapse, post-synaptic membrane become more permeable to both sodium ions and potassium ions. Depolarisation (EPSP) is due to greater invward diffusion of sodium ions into the post-synaptic neurone than outward diffusion of potassium ions.
Action potential occurs immediately in response to a stimulus of adequate strength	There is a synaptic delay due to time taken for the release of the neurotransmitter from synaptic vesicles, and for the diffusion of the neurotransmitter across the synaptic cleft
Action potential is an all or none response	Several EPSP can be summated to reach the threshold potential
Recovery is due to repolarisation (influx of sodium ions stops and efflux of potassium ions) and sodium-potassium pumps.	Recovery requires acetylcholinesterase/enzymes, which hydrolyses acetylchoine/neurotransmitters
An absolute refractory period and relative refractory period exist	No refractory period period but synapse can be fatigued due to shortage of neutrotransmitter
Speed of transmission is affected by diameter of axon, presence or absence of myelin sheath and temperature	Synaptic transmission can be affected by drugs, eg. curare. Curare binds to the receptor sites for acetylcholine
Information is transmitted over a distance of one metre	Information is tranmitted over a distance of 20nm

9(c) Explain how the temporal and spatial summation of various stimuli can result in a coordinated response at the postsynaptic neuron. [4]

Temporal (2 max)

1 refers to the summation of EPSPs produced by repeated stimulation of only **ONE** presynaptic neurone (at high frequency);
2 if more neurotransmitters are released into the cleft before the first EPSP is destroyed;
3 additive effects of several EPSPs may exceed threshold potential to result in an **AP**.

Spatial (2 max)

4 refers to the summation of EPSPs & IPSPs produced by several different pre-synaptic neurones terminating on a single post-synaptic neurone;
5 when EPSPs arrive at the same time, combined effects will result in an AP;
6 occurs when several different presynaptic terminals release neurotransmitters simultaneously;